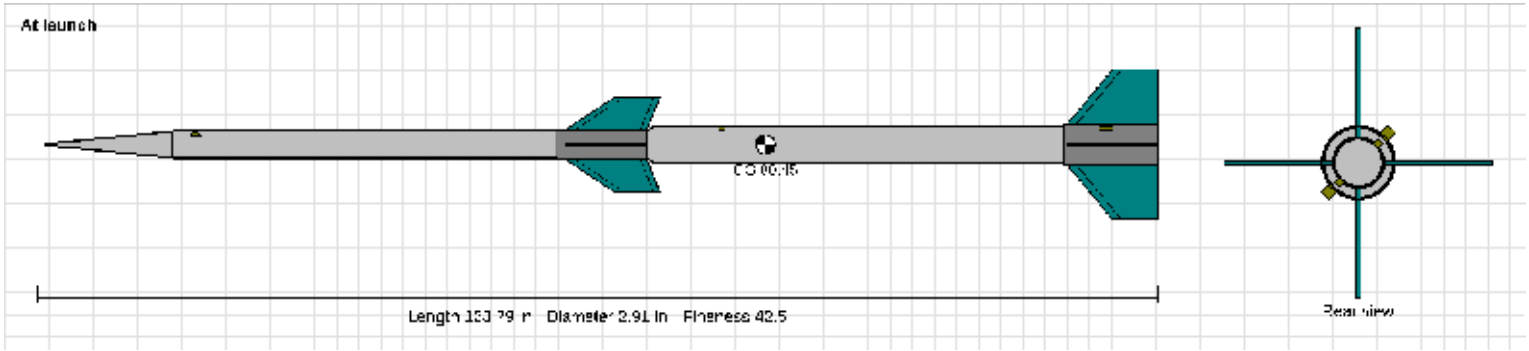
PANDORA - AMA_CONFIG_15 - Results



PANDORA - AMA_CONFIG_15
Nose radius for heating: 0.1 in

	nominal	
Rail exit speed	115.3	ft/s
Rail exit AoA	3.7515	deg
Forecast gust	7.874	ft/s
Rail exit AoA in gust	3.9068	deg
Apogee	4.0891e+05	ft
Time to apogee	173.03	s
Max speed	6080.6	ft/s
Max Mach	6.228	
Max dynamic pressure	8861.4	psf
Peak heat flux	951.15	BTU/ft²/s
Heat load	6535.4	BTU/ft²
Tilt at ignition	3.2311	deg
Altitude at ignition	31485	ft
Flight time	567.97	s
Landing east	-2.96	mi
Landing north	-9.6118	mi
Impact speed	36.042	ft/s
Launch mass	71.337	lb, CG 80.45 in from the nose tip
Min static margin	-8.5437	cal (while loaded)
Max angle of attack	0.40602	deg (while loaded)

Sustainer ignited: yes
Launch cylinder, 17 nmi: the nominal flight lands inside

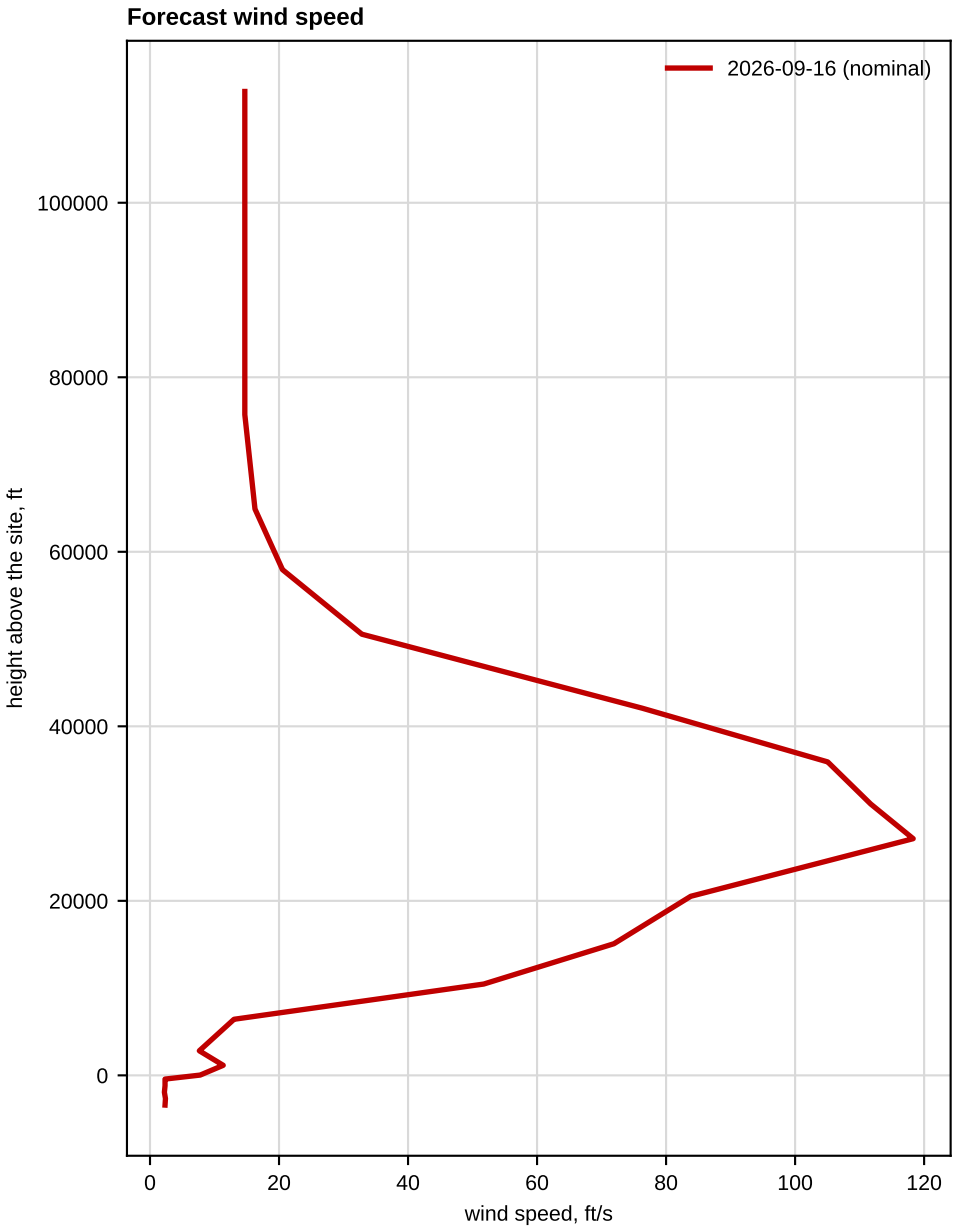
Events

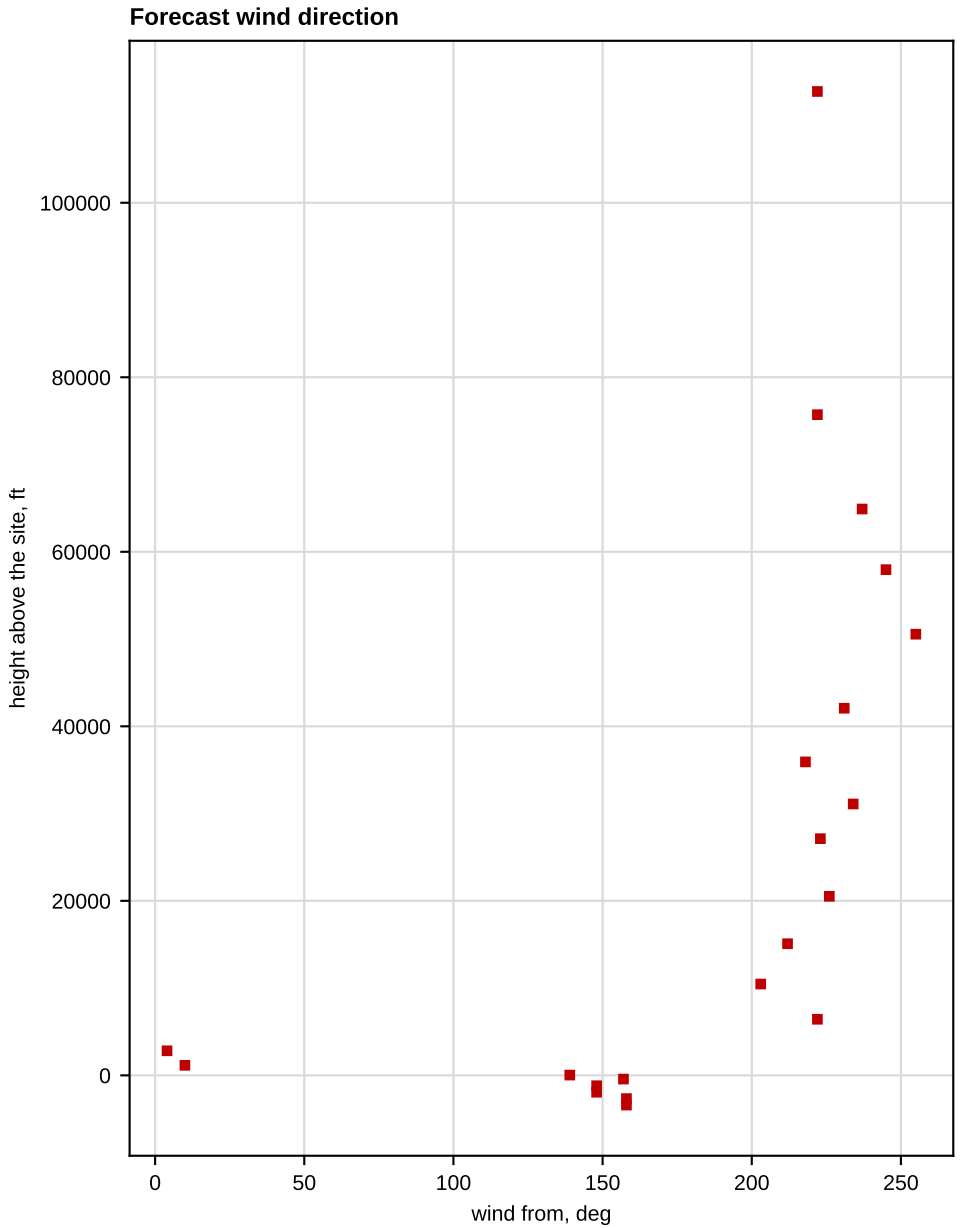
3.43 s	Booster burnout, separation
18.43 s	Sustainer ignition
21.52 s	Sustainer burnout
173.03 s	Apogee
173.03 s	Apogee chute
513.80 s	Main chute

Nose cone and fins, each alone in DATCOM (nominal flight, sustainer after separation)
  Peak nose cone drag 31.18 lbf at 3.43 s, Mach 2.77; CP 9.613 in from the nose tip
  Peak fin set drag 50.32 lbf (12.58 per fin) at 3.43 s, Mach 2.77; CP 63.99 in from the nose tip

PANDORA - AMA_CONFIG_15 - Weather

```
Weather
  Open-Meteo forecast at 40.8707, -119.1069, 06:00 America/Los_Angeles
  day                   surface wind         gust at the launch hour
  2026-09-16      2.3 ft/s from 158 deg    7.9 ft/s from 139 deg at 06:00   nominal flight
  Each day is a whole forecast wind profile; the gust is the 10 m forecast for the launch hour, used only for the rail exit AoA in gust.
```





PANDORA - AMA_CONFIG_15 - Inputs

```
[Case]
  Units               IN
  Analyze             STACK

[Nose]
  Shape               CONICAL
  Length              15
  PowerExponent       0.5
  TipRadius           0.1
  FlatTip             false

[Body]
  Diameter            2.91
  Length              52.29
  BoattailLength      0
  BoattailDiameter    3
  NozzleExitDiameter  2.5

[Fins]
  Count               4
  RootChord           9
  TipChord            5
  Span                3.65
  Sweep               5.5
  AftOffset           0
  Thickness           0.147
  Section             CHAMFER
  LEChamfer           1.019
  TEChamfer           0.7

[FinCan]
  Enabled             true
  Diameter            3.15
  Length              10.05
  AftOffset           0
  FrontBevel          1.09

[Booster]
  Enabled             true
  Diameter            4.25
  Length              56.5
  Insertion           0
  TransitionLength    1
  NoseCone            false
  NoseShape           OGIVE
  NoseLength          8
  BoattailLength      0
  BoattailDiameter    0.01
  NozzleExitDiameter  3.5

[BoosterFins]
  Count               4
  RootChord           10
  TipChord            5
  Span                6
  Sweep               5
  AftOffset           0
  Thickness           0.187
  Section             CHAMFER
  LEChamfer           1
  TEChamfer           0.75

[BoosterFinCan]
  Enabled             true
  Diameter            4.5
  Length              10.5
  AftOffset           0
  FrontBevel          0.5

[Protuberance1]
  Type                FAIRING
  Stage               SUSTAINER
  X                   17
  Count               2
  Length              1
  Width               0.4
  Height              0.4
  Standoff            0

[Protuberance2]
  Type                BUTTON
  Stage               BOOSTER
  X                   8
  Count               2
  Length              1
  Width               0.65
  Height              0.65
  Standoff            0

[Protuberance3]
  Type                LUG
  Stage               BOOSTER
  X                   50
  Count               2
  Length              1.5
  Width               0.65
  Height              0.65
  Standoff            0

[Flight]
  Altitude            4000
  ClipFinsToShock     false
  TurbulentBoundaryLayertrue

[Motors]
  Sustainer           N3364.eng
  Booster             O8220.eng

[Site]
  Weather             OPENMETEO
  Latitude            40.8707
  Longitude           -119.107
  Elevation           3701
  Date                2026-09-16
  Time                06:00
  Timezone            America/Los_Angeles
  WindSpeed           0
  WindDirection       0
  CylinderRadius      17

[Launch]
  RailLength          15
  Angle               90
  Heading             0
  Coast               15
  MaxTilt             10
  MinIgnitionAltitude 0
  MaxTime             1000

[Recovery]
  ApogeeChuteDiameter 8
  ApogeeChuteCd       0.8
  MainChuteDiameter   28
  MainChuteCd         1.8
  MainChuteAltitude   2000

[Thermal]
  IncludeDescent      false

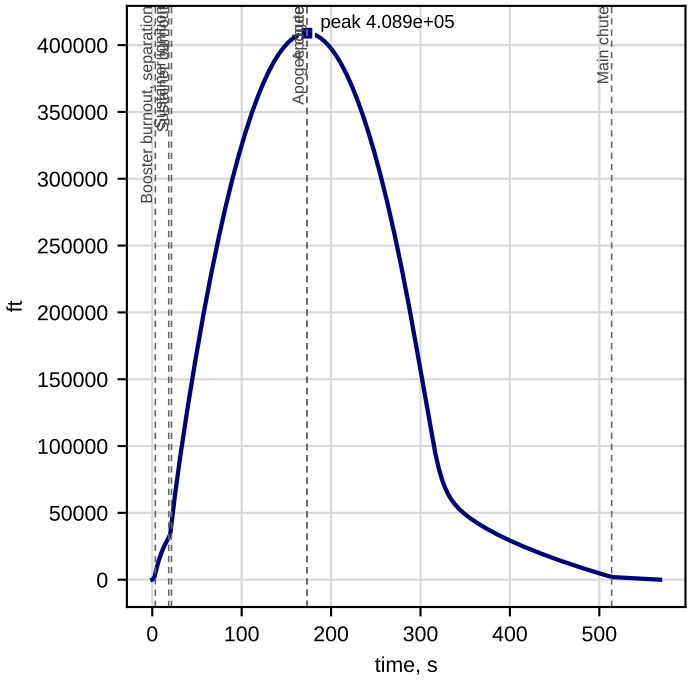
[DrySustainer]
  Mass                10.092
  CG                  41.9
  Inertia             4500
  MotorOffset         2.6

[DryStack]
  Mass                34.387
  CG                  85
  Inertia             30212.9
  MotorOffset         3.5

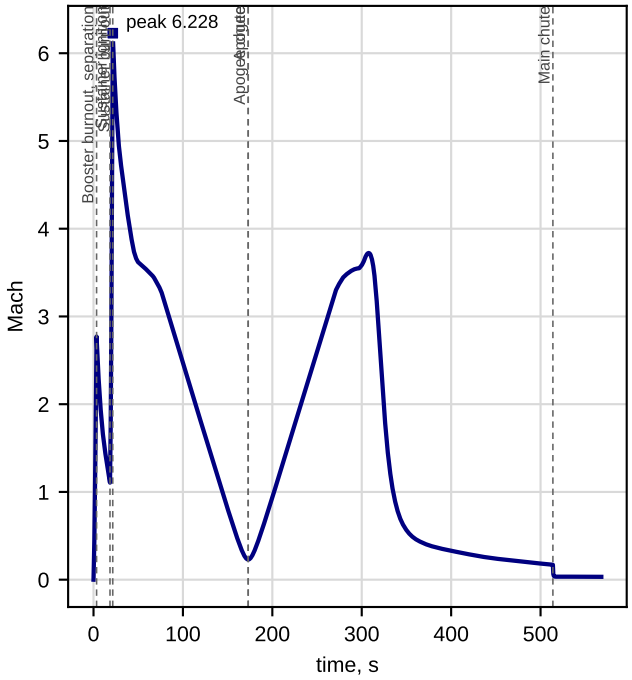
[MonteCarlo]
  Runs                0
  Seed                0
  AngleStd            0.5
  HeadingStd          180
  CoastStd            0.5
  BoosterImpulseStd   2
  SustainerImpulseStd 2
  BoosterMassStd      2
  SustainerMassStd    2
  WindSpeedStd        0
  WindDirectionStd    0
  WindDays            7
  MaxDayWind          10
  SkipDays
```

AMA_CONFIG_15 - Flight

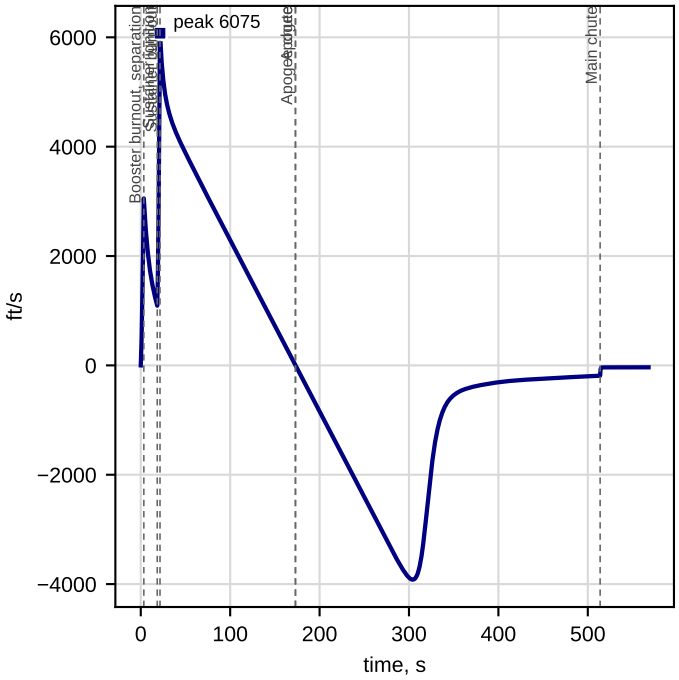
Altitude



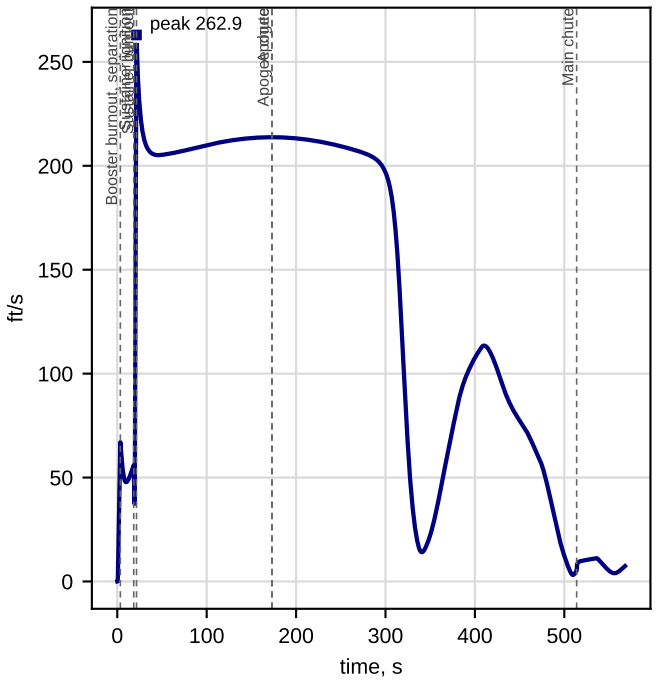
Mach number



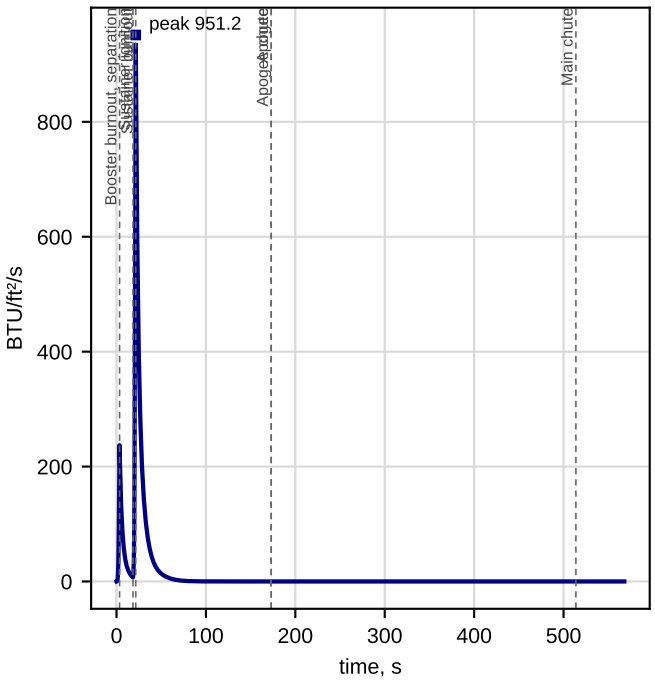
Vertical velocity



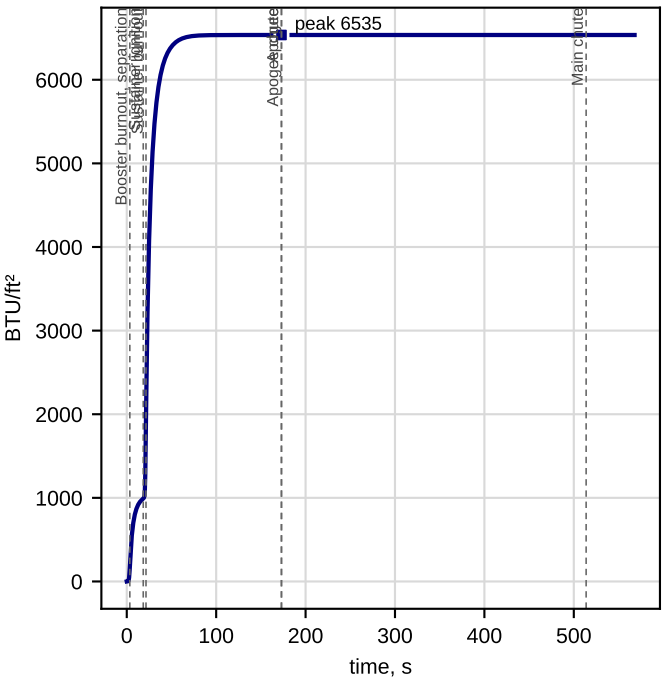
Horizontal velocity



Stagnation heat flux

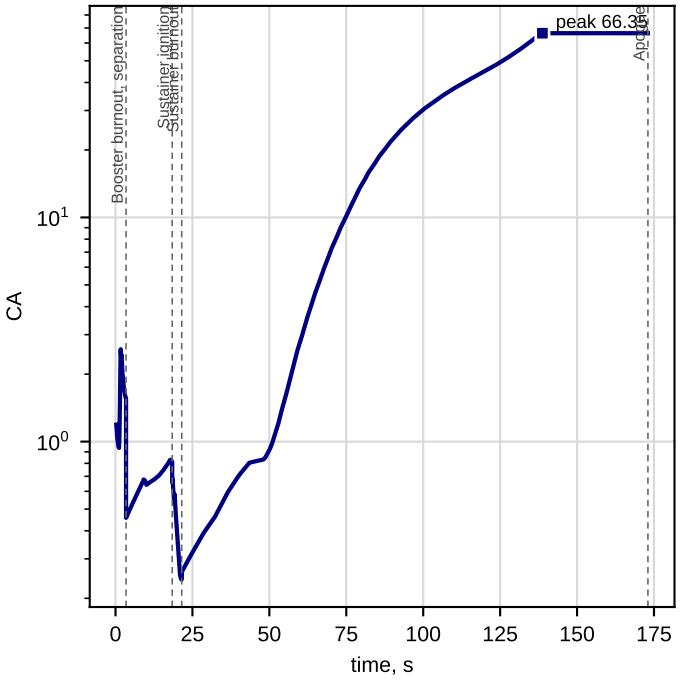


Integrated heat load

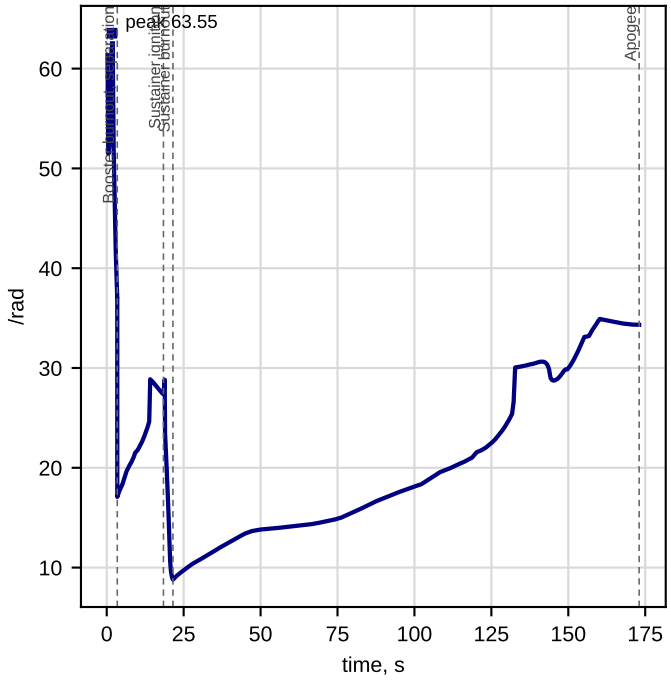


AMA_CONFIG_15 - DATCOM aerodynamics along the flight (launch to apogee, Mach 0.1 and up)

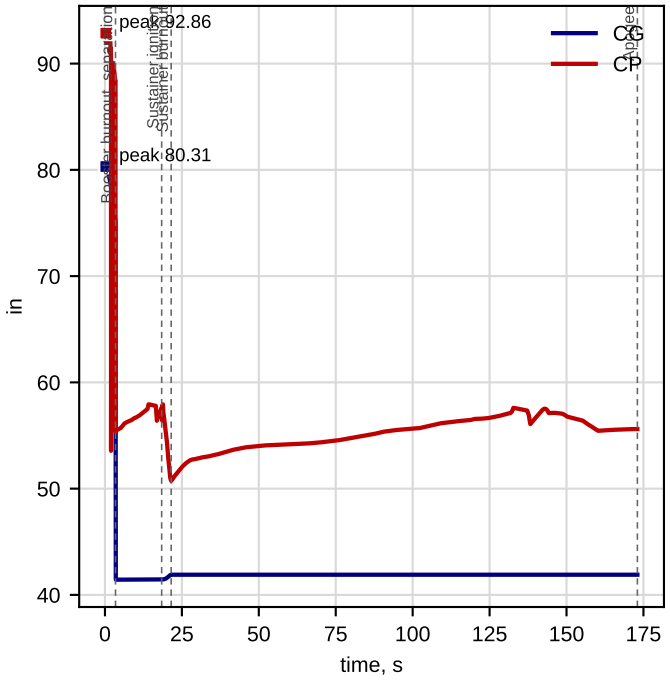
Axial force coefficient CA (log scale)



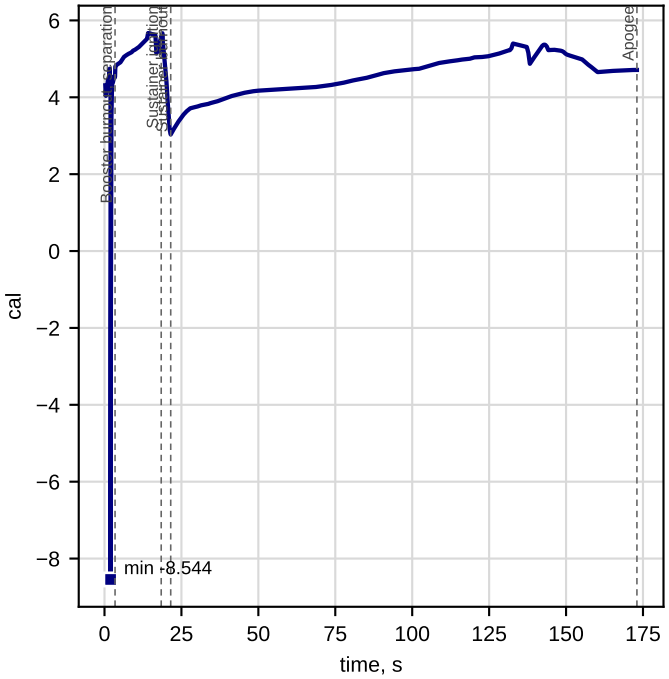
Normal force slope CNα



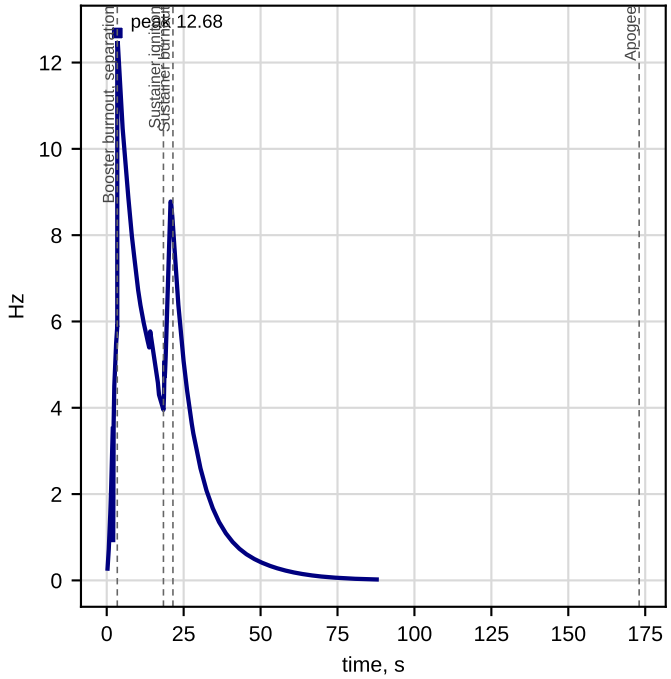
CG and centre of pressure



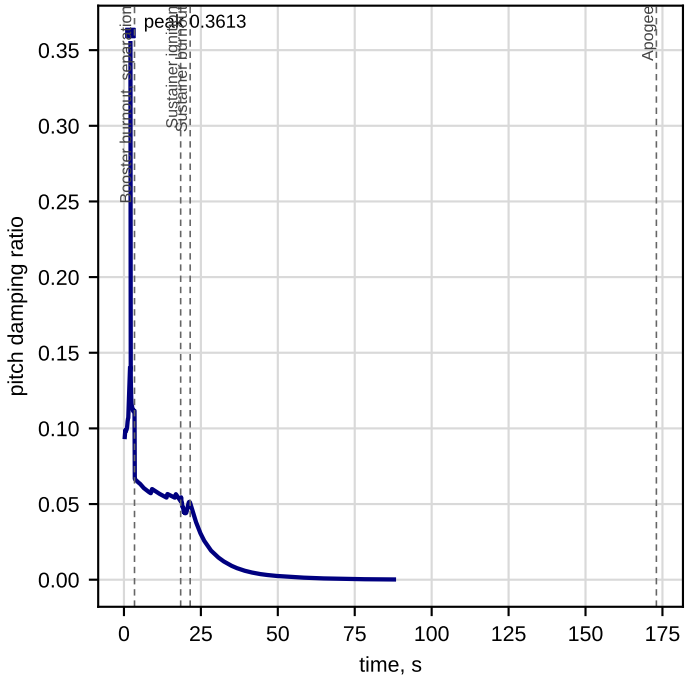
Static margin



Pitch natural frequency



Pitch damping ratio



AMA_CONFIG_15 - Ground track

